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Project title : Obstacle avoiding and line following car

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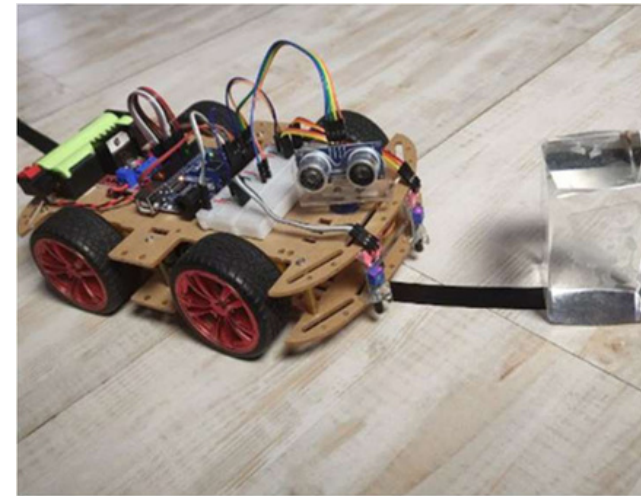
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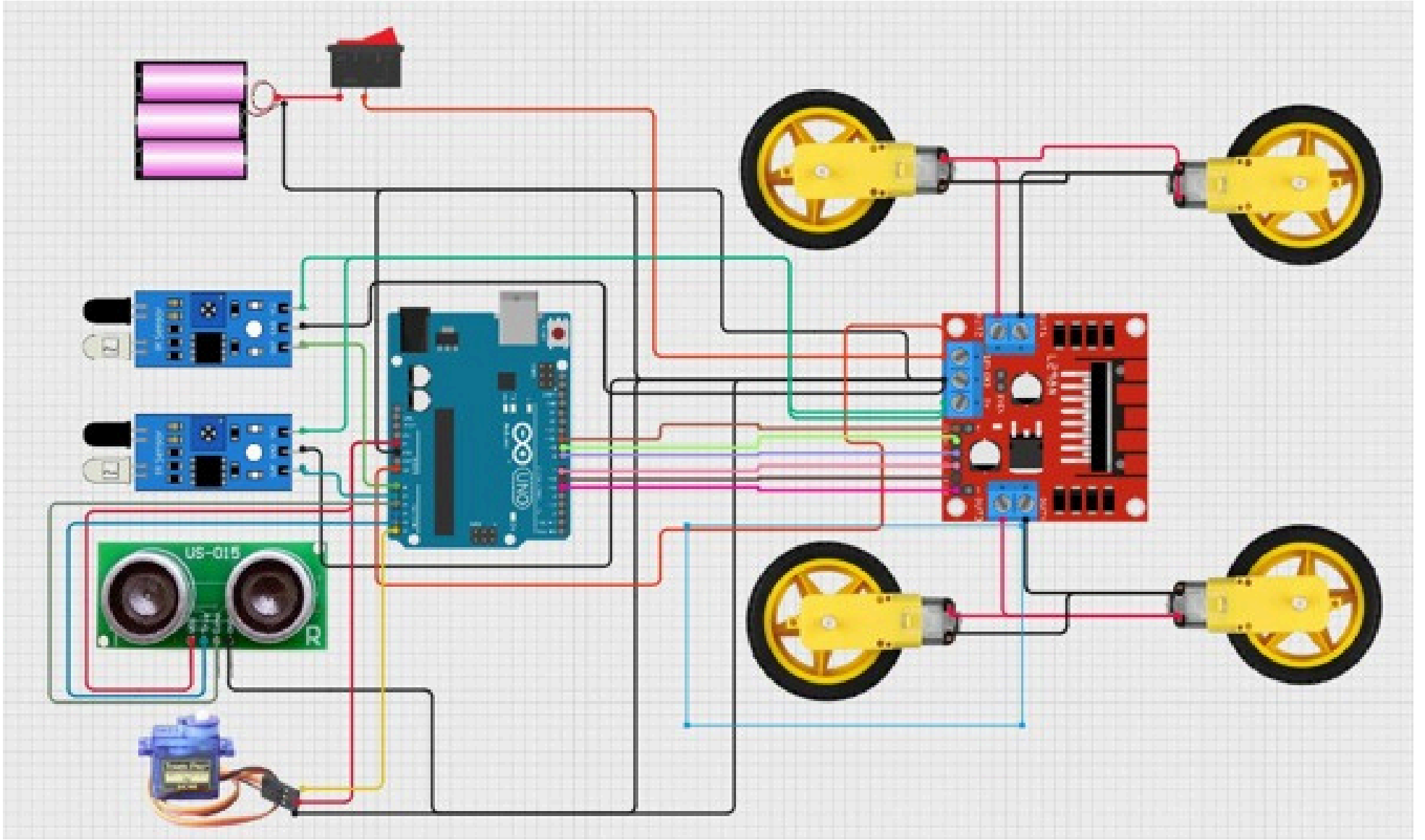
Introduction:

- The increasing role of automation and robotics in transportation and logistics .
- Introduction to embedded systems and Arduino platforms in low-cost smart vehicle development .
- Purpose of developing a hybrid car capable of both obstacle avoidance and line following.
- The system provides an effective platform for learning and experimenting with real-time decision-making and autonomous navigation using affordable components.

Objective:

- To design and build an autonomous car that can:
Follow a predefined line using sensors.
- Detect and avoid obstacles in its path.
- Ensure smooth navigation and collision-free movement in dynamic environment
- Be controlled wirelessly .
- Promote hands-on learning of embedded systems and automation logic





Work Plan:

- Literature Study & Component Selection
- Circuit Design & Simulation
- Hardware Assembly
- Coding & Integration
- Testing & Debugging
- Documentation & Submission

Outcome:



- The car follows a marked line on the floor smoothly.
- When an obstacle appears on the path, the car:
 1. Detects it using ultrasonic/IR sensors.
 2. Stops, avoids, or reroutes around the obstacle.
 3. Returns to the line and continues following the path.
- Demonstrates autonomous decision-making without manual control.
- Provides a practical prototype for smart vehicles, warehouse robots, and industrial automation.



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